Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void fun(int n){
    int counter = 0;
    int c = 0; counter++;
    for(int i=n/2; i<n; i++){
        counter++;
        for(int j=1; j<n; j *= 2){
            counter++;
            for(int k=1; k<n; k*=2){
                counter++;
                c++; counter++;
            }
            counter++; //false for k loop
        }
        counter++; //false for j loop
    }
    counter++; //false for i loop
    printf("%d",counter);
}

int main(){
    int n;
    scanf("%d",&n);
    fun(n);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course